

ITCS 2026

Debordering Closure Results in Determinantal and Pfaffian Ideals

Anakin Dey & Zeyu Guo

Ohio State University

Algebraic Circuits

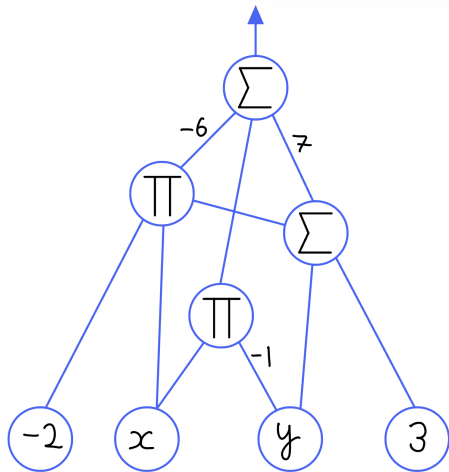
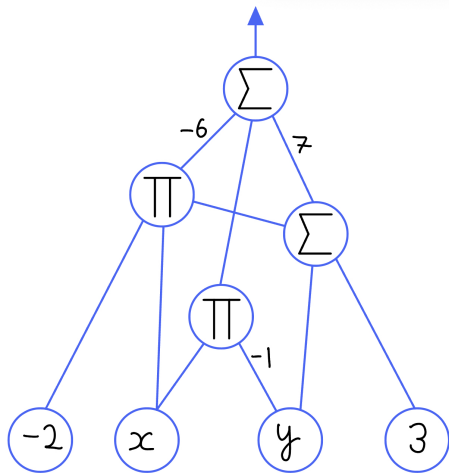


Figure: $-6(-2x(y+3)) - xy + 7(x+3)$

Algebraic Circuits



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

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Algebraic Circuits with Oracles

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Instead of just allowing Σ and Π gates, we can take any polynomial $f(\bar{x})$ and use this as a *f -oracle* gate.

For example, if $f(\bar{x}) = r(\bar{x}) \cdot g(\bar{x})$, we can ask about the circuit complexity for g if we allow an f -oracle gate.

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A subset $I \subseteq \mathbb{F}[\bar{x}]$ is called an *ideal* if it satisfies

1. $f(\bar{x}), g(\bar{x}) \in I \implies f(\bar{x}) + g(\bar{x}) \in I$
2. $f(\bar{x}) \in I, r(\bar{x}) \in \mathbb{F}[\bar{x}] \implies r(\bar{x}) \cdot f(\bar{x}) \in I$

Circuit Complexity for Ideals

Theorem (Hilbert's Basis Theorem)

If $I \subseteq \mathbb{F}[\overline{x}]$ is an ideal, then there exist finitely many polynomials $g_1(\overline{x}), \dots, g_k(\overline{x}) \in I$ which *generate* I .

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Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial, so $I = \langle g \rangle$.
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Question

Suppose $I = \langle g \rangle$. Let $f \in I$. How does the complexity of f compare to the complexity of g ?

In other words, if g is a factor of f and f has a small circuit, does g have a small circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

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Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *approximately computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

By approximate computation, we mean the circuit computes a polynomial $h(\bar{x}) \in \mathbb{F}(\varepsilon)[\bar{x}]$ such that $h(\bar{x}) - g(\bar{x}) \in \varepsilon \mathbb{F}[[\varepsilon]][\bar{x}]$.

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Can we *deborder* this result, that is can we remove this ε approximation?

Determinantal Ideals

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Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

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Conjecture ([Gro, Conjecture 6.3])

For every nonzero polynomial f in $I_{n,n,n/2}$, there is a constant-depth f -oracle algebraic circuit of size $\text{poly}(n)$ which computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *approximately computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

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Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

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- We also prove analogous results for *Pfaffian* ideals in the case that our matrix of variables X is skew-symmetric ($x_{j,i} = -x_{i,j}$).
- We need that our fields have size $\text{poly}(\deg(f), n, m)$ as well as characteristic greater than $\deg(f)$.

Bideterminants

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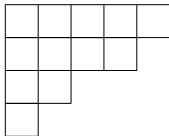


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A *Young Tableau* of shape σ is a filling of the squares in the diagram for σ using positive integers.

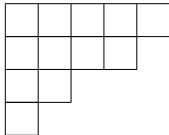


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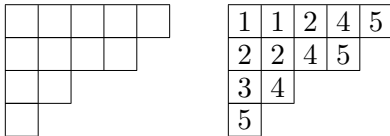


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Such a filling is *conjugate semistandard* if it is strictly increasing down columns and weakly increasing along rows.

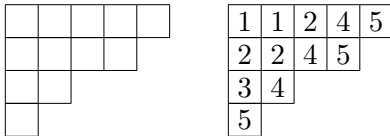


Figure: Diagram for $\sigma = (5, 4, 2, 1)$ and a Young Tableau of shape σ

Bideterminants

We have *canonical* semistandard tableau.

1	2	3	4	5
1	2	3	4	
1	2			
1				

Figure: Canonical tableau of shape $\sigma = (5, 4, 2, 1)$

We will notate the canonical semistandard tableau of shape σ by K_σ .

Bideterminants

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$ and $(S \mid T)$ is the *bitableau*

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right),$$

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then the *bideterminant* $(S \mid T)(X)$ is given by

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A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

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Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X),$$

where A and B are *conjugate semistandard*.

Homogenization

Definition

Consider a degree d polynomial $f \in \mathbb{F}[\bar{x}, t]$. Then we have polynomials $\text{coeff}_{t^0}(f), \dots, \text{coeff}_{t^d}(f) \in \mathbb{F}[\bar{x}]$:

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This circuit is computing via interpolation at $d + 1$ *distinct* points in $\mathbb{F}$. Furthermore, if the top gate of the circuit for f is a Σ -gate, then $\text{coeff}_{t^i}(f)$ can be computed with depth Δ .

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Then with probability at least $1 - \varepsilon$, there is a *unique* form in $\mathcal{L}$ attaining the minimum value.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $f(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in f is at most K :

$$f(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

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Then with probability at least $1 - \varepsilon$, there is a *unique* monomial of f with which attains the minimum degree in w under this morphism.

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Using the Isolation Lemma and Homogenization, we extract a *single* monomial and compute this monomial with a small circuit.

Proof Outline

Since we are starting with $f \in I_{n,m,r}^{\det}$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X)$$

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Our goal will be to use $f(X)$ as an oracle to compute a canonical bideterminant.

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- The terms indexed by $\ell \in B$ are “garbage terms” produced by this linear map.

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k}(K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} b_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell}(S_\ell \mid T_\ell)(X)$$

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} b_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} (S_\ell \mid T_\ell)(X)$$

We do another linear transformation to get an expression of the form

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} v^{D_k} (K_{\sigma_k} \mid K_{\sigma_k})(X) + \text{terms of lower degree in } v$$

Proof Outline

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Homogenization $\implies$ f -oracle circuit for

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

All of these terms now have the “same total degree” so we cannot homogenize to keep a single one.

Proof Outline

$$\sum_{k \in A} a_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

All of these terms now have the “same total degree” so we cannot homogenize to keep a single one.

Apply a random map

$$\lambda_{a,b} \mapsto w^{t_{a,b}}$$

$$\xi_{i,j} \mapsto w^{t_{i,j}}$$

$$\sum_{k \in A} a_k w^{\bar{t}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k w^{\bar{t}_k}(K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Proof Outline

$$\sum_{k \in A} a_k w^{\bar{t}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Isolation Lemma $\implies$ there is a choice of the $t_{i,j}$ not too large so that every term has *distinct* degree in w .

Proof Outline

$$\sum_{k \in A} a_k w^{\bar{t}_k} (K_{\sigma_k} \mid K_{\sigma_k})(X)$$

Isolation Lemma $\implies$ there is a choice of the $t_{i,j}$ not too large so that every term has *distinct* degree in w .

We can then use Homogenization again to extract a single term.

$$a(K_{\sigma} \mid K_{\sigma})(X)$$

Thank You!

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